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Copper catalysis for highly selective aerobic oxidation of alcohols to aldehydes/ketones†

Di Zhai^a and Shengming Ma^{*,b}

Oxidation of alcohols to aldehydes/ketones is a transformation of fundamental importance. Usually stoichiometric amounts of oxidants are required for these types of reactions. Although aerobic oxidations have been studied, there exist challenges: complicated additives, toxic or expensive catalyst(s), a large amount of base or molecular sieves, limited substrate scope, isomerization of $Z\ C=C$ bonds, *etc.* Herein, a general and practical room temperature aerobic oxidation of different types of alcohols including allenols, alkenols, propargylic alcohols, benzylic alcohols, and aliphatic alcohols in MeCN (acetonitrile) or DCE (1,2-dichloroethane) using readily available and cheap $Cu(NO_3)_2 \cdot 3H_2O$ and TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy) or 4-HO-TEMPO as the catalysts forming aldehydes or ketones with an excellent selectivity has been developed. Some thirty to forty mmol scale reactions have been demonstrated.

Introduction

Aldehydes and ketones are very common and useful chemicals for organic chemistry, cosmetics, medicinal chemistry, and materials science. They are usually synthesized *via* oxidation of alcohols both in academic laboratories and industries. Traditional protocols for the oxidation of alcohols require at least a stoichiometric amount of certain expensive oxidants such as MnO_2 , CrO_3 , PCC, PDC, IBX, *etc.*, which would yield an equal amount of waste, resulting in a serious problem for isolation of the products and a big burden to the environment.¹ Therefore, much attention has been focused on the eco-friendly aerobic oxidation of alcohols.^{1,2}

In 1981, Gilbride and his coworkers^{3a} found that there is a high level of copper in the recovered product of the protein from the serum of a horseshoe crab. Later, Linzen and his coworkers^{3b} found in the hemocyanins of arthropods and mollusks two copper atoms bound directly to side chains at the site of oxygen binding while a single iron was contained in the heme of hemoglobin. Thus, copper is the metal of choice in nature to carry oxygen. In fact, Semmelhack, Stahl, Iwabuchi, Markó, and their coworkers^{4–6} have made very important contributions in the area of using copper as the

catalyst for aerobic oxidation of alcohols. In addition, a catalytic amount each of $Cu(MeCN)_4PF_6$, DBED, and DMAP⁷ or $CuCl$ combined with highly toxic OsO_4 ⁸ may also catalyze this reaction, however, a large amount of 4 Å molecular sieves was required, which makes the reaction non-practical in large-scale productions (Scheme 1). Complicated additives, toxic or expensive catalyst(s), high temperature, a large amount of base or molecular sieves, limited substrate scope (for further examples, see also: Tables 3 and 4), serious isomerization of $Z\ C=C$ bonds,^{5c} *etc.* call for the development of new catalytic recipes. Thus, cost-effective highly selective aerobic oxidation of alcohols with a wide scope and academic/industrial potential at room temperature is of high current interest. The common approach is usually the development of ligands or more expensive copper catalysts with more complicated amine *N*-oxyls.^{4,5} We tried the copper anion effect and observed that these above-mentioned challenges may be addressed by applying copper nitrate and TEMPO as catalysts, oxidizing a very wide scope of alcohols to aldehydes or ketones with an excellent selectivity (Scheme 1).

Results and discussion

At the beginning of our study, we selected 1-phenyl-2-heptyn-1-ol **1a** as the model starting material to optimize the reaction conditions of the copper-catalyzed aerobic oxidation reaction. First, different solvents were screened; when NMP, cyclohexane and $MeNO_2$ were used as solvents (Table 1, entries 1, 2 and 4), 56–84% of **1a** was recovered. Ethers such as dioxane, THF, and Et_2O only gave 42–85% yield of **2a** (Table 1, entries 3, 5, and 6). DCM and toluene were also examined with 91% and 92% yield

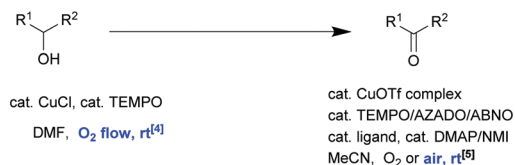
^aResearch Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Lu, Shanghai 200433, P. R. China.

E-mail: dxdzwl@sina.com

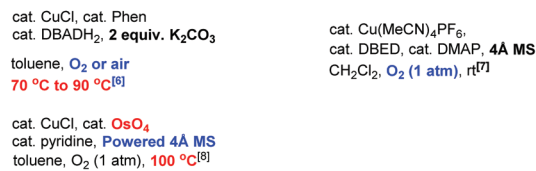
^bState Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China. E-mail: masm@sioc.ac.cn

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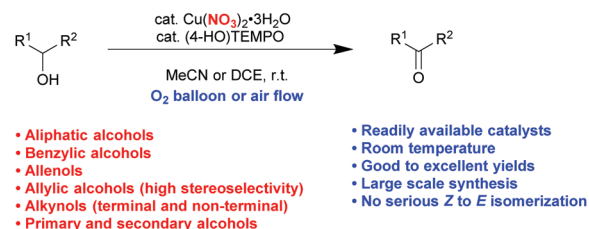
Typical known approach 1-improved with more expensive Cu catalyst, ligand, and amine *N*-oxyl:



Typical known approach 2-modified with stoichiometric base, more expensive Cu catalyst, ligand, toxic OsO₄, or MS:



This work-the anion effect:



Scheme 1 Copper-catalyzed aerobic oxidation of alcohols to aldehydes/ketones.

Table 1 The effect of solvent in the aerobic oxidation of **1a**^a

Entry	Solvent	Yield of 2a ^b (%)	Recovery of 1a ^b (%)
1	NMP	11	84
2	Cyclohexane	12	76
3	Dioxane	42	50
4	MeNO ₂	37	56
5	THF	49	46
6	Et ₂ O	85	12
7	DCM	91	5
8	Toluene	92	8
9	DCE	99	0
10	MeCN	99	0

^aThe reaction was conducted using alcohol (0.4 mmol), Cu(NO₃)₂·3H₂O (0.04 mmol), TEMPO (0.04 mmol), and solvent (1.6 mL) with an oxygen balloon at 25 °C. ^bDetermined by ¹H NMR analysis with dibromomethane as the internal standard.

of **2a**, respectively (Table 1, entries 7 and 8). Finally, DCE and MeCN turned out to be the best solvents, both providing **2a** in 99% yield with no recovery of **1a** (Table 1, entries 9 and 10).

Based on the results in Table 1, MeCN was selected as the best solvent to investigate the effect of different copper salts. Cu(MeCN)₄PF₆^{5b} did not work well (Table 2, entry 1). Other common copper salts such as CuCl,^{4f} CuBr₂,^{5a} CuI and

Table 2 The effect of copper salts^a

Entry	Cu salt	Yield of 2a ^b (%)	Recovery of 1a ^b (%)
1	Cu(MeCN) ₄ PF ₆	6	90
2	CuCl	9	91
3	CuBr ₂	12	84
4	CuI	19	78
5	Cu(OAc) ₂	43	57
6	Cu(NO ₃) ₂ ·3H ₂ O	96	0
7	Without Cu(NO ₃) ₂ ·H ₂ O	1	94
8	Without TEMPO	3	94

^aThe reaction was conducted using alcohol (0.4 mmol), copper salt (0.04 mmol), TEMPO (0.04 mmol), and MeCN (1.6 mL) with an oxygen balloon at 25 °C. ^bDetermined by ¹H NMR analysis with dibromomethane as the internal standard.

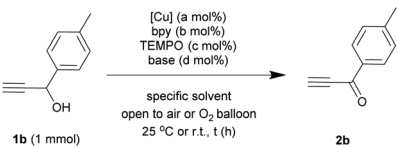
Cu(OAc)₂^{4d} only gave 9%–43% yields of **2a** (Table 2, entries 2–5). Control experiments disclosed the importance of Cu(NO₃)₂·3H₂O and TEMPO for this transformation (Table 2, entries 7 and 8).

Control experiments show that terminal propargylic alcohol, 1-(4-methylphenyl)-2-propyn-1-ol **1b**, could not be transformed into the corresponding ketone **2b** by the most popular copper catalytic systems,^{4f,5a,b} most probably due to the basic nature of the reaction conditions (Table 3, entries 1–3). The current method could afford 1-(4-methylphenyl)-2-propyn-1-one **2b** in 94% NMR yield (Table 3, entry 6). Since terminal alkynones are a type of very useful compounds in organic synthesis, we demonstrated the scope carefully. Besides 4-methylphenyl, other electron-withdrawing and electron-donating group substituted phenyl groups, thienyl and even *n*-C₉H₁₉ substituted terminal propargylic alcohols provided the desired alkynones **2b–2h** in good to excellent yields (Table 4, entries 1–7).

We also demonstrated the scope of the oxidation reaction of non-terminal propargylic alcohols to the corresponding aldehydes or ketones. When R² is Ph, different groups including *n*-Bu, cyclopropyl, Ph and TMS may be introduced as the R¹ group with excellent yields (Table 5, entries 1–4, **2a** and **2i–2k**). R¹ may also be *n*-Bu with R² being different electron-withdrawing and electron-donating group substituted phenyl groups, 2-furyl and *n*-Bu, providing the desired ketones **2l–2t** in good to excellent yields (Table 5, entries 5–13). When R¹ is aryl and R² is replaced by cyclohexyl, naphthyl and proton, the desired ketones or aldehydes can also be provided in good to excellent yields (Table 5, entries 14–17, **2u–2x**). The tolerance of synthetically useful functional groups is highlighted in red, demonstrating the potential of this methodology (Tables 4 and 5).

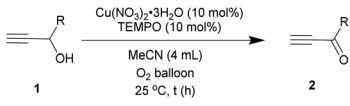
We are also pleased to observe that this protocol may be used in the oxidation of allylic alcohols. Besides good to excellent yields of these reactions, the isomerization of the C=C bond is very limited for both primary alcohols and secondary alcohols (Scheme 2, eqn (1)–(5), **4a–4e**).^{5c}

Table 3 Control experiment of aerobic oxidation of terminal propargylic alcohol **1b**

								
Experimental conditions								
Entry	[Cu] (a)	b	c	Base (d)	t (h)	Solvent	Yield of 2b ^a (%)	Recovery of 1b ^a (%)
1 ^b	Cu(MeCN) ₄ PF ₆ (5)	5	5	NMI (10)	24	MeCN	0	23
2 ^b	CuBr ₂ (5)	5	5	<i>t</i> -BuOK (5)	24	MeCN/H ₂ O (2 : 1)	0	33
3 ^b	CuCl (5)	—	5	DMAP (10)	24	H ₂ O	0	83
4 ^b	Cu(NO ₃) ₂ ·3H ₂ O (10)	—	10	—	22	MeCN	81	0
5 ^c	Cu(NO ₃) ₂ ·3H ₂ O (10)	—	10	—	6	DCE	46	48
6 ^c	Cu(NO ₃) ₂ ·3H ₂ O (10)	—	10	—	6	MeCN	94 (91 ^d)	0

^a Determined by ¹H NMR analysis with dibromomethane as the internal standard. ^b The reaction system was open to air. ^c The reaction was conducted with an oxygen balloon. ^d The number in parenthesis represents the isolated yield of **2b**.

Table 4 Room temperature aerobic oxidation of terminal 2-alkynols to the corresponding ketones^a

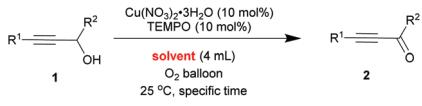
			
Entry	R (1)	t (h)	Isolated yield of 2 ^a (%)
1	4-MeC ₆ H ₄ (1b)	6	91 (2b)
2	2-MeOC ₆ H ₄ (1c)	12.5	84 (2c)
3	3-MeOC ₆ H ₄ (1d)	13	89 (2d)
4	4-MeOC ₆ H ₄ (1e)	6	90 (2e)
5	4-MeO ₂ CC ₆ H ₄ (1f)	12.5	90 (2f)
6	2-Thienyl (1g)	6	87 (2g)
7	<i>n</i> -C ₉ H ₁₉ (1h)	8	89 ^b (2h)

^a The reaction was conducted using alcohol (1.0 mmol), Cu(NO₃)₂·3H₂O (0.1 mmol), TEMPO (0.1 mmol) and MeCN (4 mL) with an oxygen balloon at 25 °C. ^b DCE was used as a solvent.

Benzylic alcohols, aliphatic alcohols, and allenols could also be converted into the corresponding aldehydes and ketones by only replacing MeCN with DCE (Scheme 3, eqn (1)–(7), **6a**, **6b**, **6c**, **8a**, **8b**, **10a**, and **10b**).

To further demonstrate the potential and efficiency of this catalytic system, we conducted some reactions on scales practical in an academic laboratory. The reaction may be conducted smoothly on a 30 mmol scale with 2.5 mol% catalyst loading, giving alkynones **2a** in 98% isolated yield (Scheme 4, eqn (1)). Three types of solutions have been provided: using a bag of air (size: 48 L, oxygen bags used in hospitals) for 1.5 h first with the subsequent supply of the consumed oxygen with a bag of pure O₂ (commercial size: 2 L, may be expanded to 5 L) may efficiently avoid the unsafe high concentration of pure O₂ over MeCN, which is critical for industrial scale production⁹ (Scheme 4, eqn (2)). The oxidation of 1-(2-thienyl)-2-propyn-1-ol **1g** was also conducted on a 40 mmol scale by applying a slow flow of air (30 mL min⁻¹),¹⁰ giving 79% isolated yield of **2g** (Scheme 4, eqn (3)).

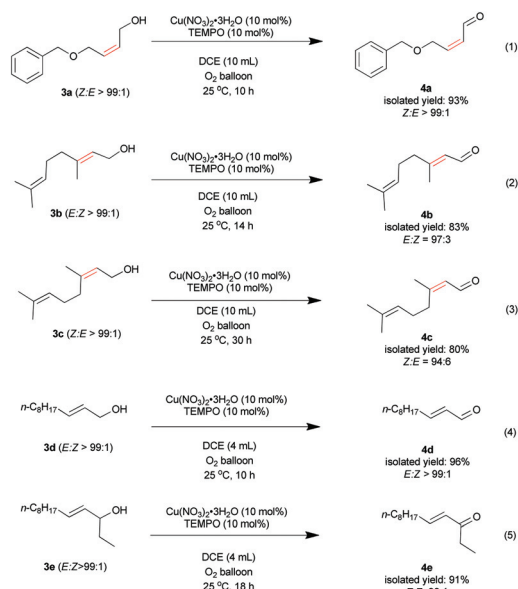
Table 5 Room temperature aerobic oxidation of non-terminal 2-alkynols to aldehydes or ketones^a

			
Entry	R ¹ /R ²	Isolated yield of 2 (%)	
		MeCN	DCE
1	<i>n</i> -Bu/Ph (1a)	96 (2a)	100 (2a)
2	Cyclopropyl/Ph (1i)	98 (2i)	96 (2i)
3	Ph/Ph (1j)	98 (2j)	— ^b
4	TMS/Ph (1k)	97 (2k)	95 (2k)
5	<i>n</i> -Bu/3-ClC ₆ H ₄ (1l)	97 (2l)	98 (2l)
6	<i>n</i> -Bu/2-BrC ₆ H ₄ (1m)	97 (2m)	98 (2m)
7	<i>n</i> -Bu/4-MeOC ₆ H ₄ (1n)	100 (2n)	97 (2n)
8	<i>n</i> -Bu/4-MeO ₂ CC ₆ H ₄ (1o)	99 (2o)	— ^b
9	Ph/2-HOC ₆ H ₄ (1p)	— (2p) ^c	81 (2p)
10	<i>n</i> -Bu/3-NCC ₆ H ₄ (1q)	92 (2q)	— ^b
11	<i>n</i> -Bu/4-O ₂ NC ₆ H ₄ (1r)	97 (2r)	— ^b
12	<i>n</i> -Bu/2-furyl (1s)	92 (2s)	95 (2s)
13	<i>n</i> -Bu/ <i>n</i> -Bu (1t)	96 (2t)	95 (2t)
14	Ph/ <i>n</i> -Bu (1u)	93 (2u)	96 (2u)
15	Ph/cyclohexyl (1v)	95 (2v)	100 (2v)
16	Ph/naphthyl (1w)	95 (2w)	95 (2w)
17	4-BrC ₆ H ₄ /H (1x)	76 (2x)	65 (2x)

^a The reaction was conducted using alcohol (1.0 mmol), Cu(NO₃)₂·3H₂O (0.1 mmol), TEMPO (0.1 mmol) and MeCN (4 mL) or DCE (4 mL) with an oxygen balloon at 25 °C, and the reaction time is shown in the ESI.† ^b Not conducted. ^c 49% yield of **2p** and 38% yield of (*Z*)-2-benzylidenebenzofuran-3(2*H*)-one were detected, which were determined by ¹H NMR analysis with dibromomethane as the internal standard.

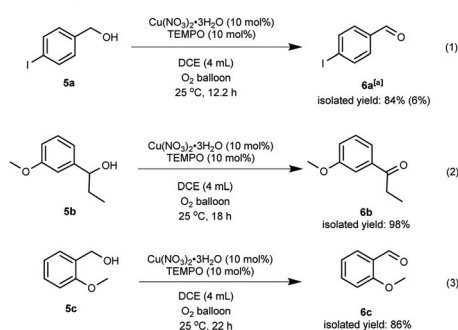
In consideration of the fact that 4-OH-TEMPO is much cheaper than TEMPO,¹¹ we also conducted the aerobic oxidation of some alcohols by using 4-OH-TEMPO (Scheme 5).

Compared to Fe(NO₃)₃·9H₂O,¹² the aerobic oxidation reaction of alkynol **1g** catalyzed by Cu(NO₃)₂·3H₂O is faster (Scheme 6). Furthermore, we observed that the addition of sodium chloride, which was applied as a co-catalyst to speed

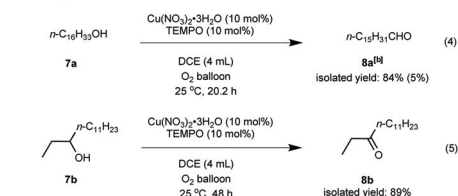


Scheme 2 Room temperature aerobic oxidation of allylic alcohols to aldehydes or ketones. The reaction was conducted using alcohol (1.0 mmol), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.1 mmol), TEMPO (0.1 mmol) and DCE with an oxygen balloon at 25 °C.

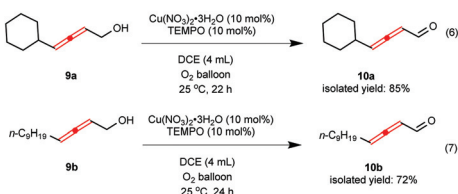
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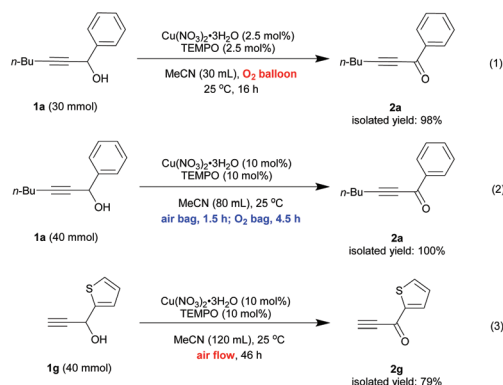
Part 2: Aliphatic alcohols



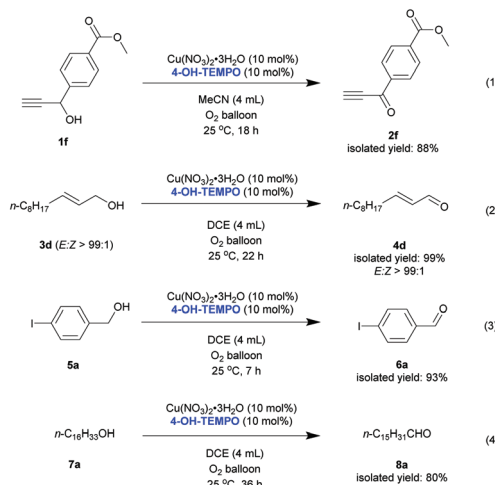
Part 3: allenols



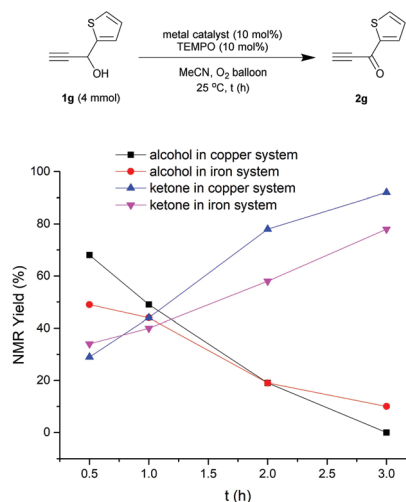
Scheme 3 Room temperature aerobic oxidation of benzylic, aliphatic, and allenyl alcohols to aldehydes or ketones. The reaction was conducted using alcohol (1.0 mmol), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.1 mmol), TEMPO (0.1 mmol) and DCE (4 mL) with an oxygen balloon at 25 °C. ^aThe number in the parentheses represents the yield of acid detected by NMR analysis. ^bThe number in the parentheses represents the recovery of alcohol detected by NMR analysis.



Scheme 4 Large-scale reactions with O_2 , air/ O_2 , or air.

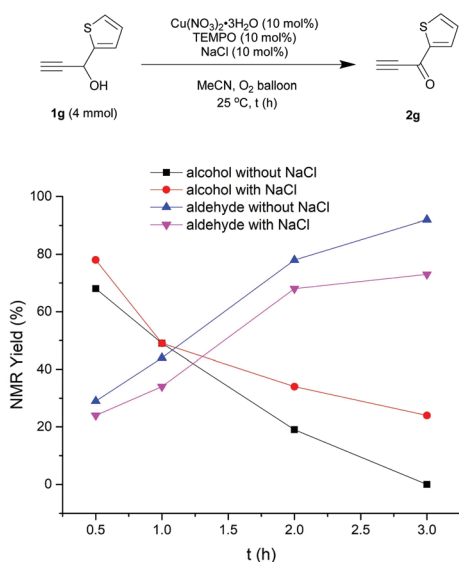


Scheme 5 The reaction with 4-OH-TEMPO instead of TEMPO. The reaction was conducted using alcohol (1.0 mmol), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.1 mmol), TEMPO (0.1 mmol) or 4-OH-TEMPO and MeCN (4 mL) or DCE with an oxygen balloon at 25 °C.



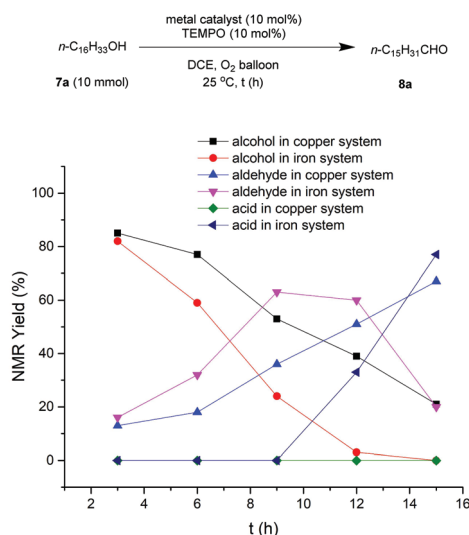
Scheme 6 Reaction profile of alkynol **1g** in the Cu or Fe system. The reaction was conducted using alcohol (4.0 mmol), metal catalyst (0.4 mmol), TEMPO (0.4 mmol) and MeCN (16 mL) with an oxygen balloon at 25 °C.

up the reaction catalyzed by $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$,¹² slowed down this copper-catalyzed oxidation system (Scheme 7).



Scheme 7 The effect of chloride. The reaction was conducted using alcohol (4.0 mmol), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.4 mmol), TEMPO (0.4 mmol) and MeCN (16 mL) with an oxygen balloon at 25 °C.

Interestingly, for primary aliphatic alcohol **7a**, the reaction with $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ is much faster with a problem of over-oxidation,^{9,10} while the reaction catalyzed by $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ demonstrated a very high selectivity for stopping at the stage of aldehydes (Scheme 8), indicating that the reaction from aliphatic aldehyde to the corresponding carboxylic acid with Cu is much slower.



Scheme 8 Reaction profile of aliphatic alcohol **7a** in the Cu- or Fe-catalyzed system. The reaction was conducted using alcohol (10.0 mmol), metal catalyst (1.0 mmol), TEMPO (1.0 mmol) and DCE (40 mL) with an oxygen balloon at 25 °C.

Conclusions

In conclusion, we have developed an eco-friendly and practical efficient aerobic oxidation of different alcohols to aldehydes/ketones using $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ and TEMPO as catalysts. This protocol tolerates most common alcohols in organic synthesis such as propargylic alcohols, allylic alcohols, allenols, benzylic alcohols and normal alkanols, providing the corresponding aldehydes/ketones in good to excellent yields with an excellent selectivity. The oxidation reaction may be easily executed on a practical scale in an academic laboratory and no over-oxidation has been observed. Further studies of this copper catalytic system including the application in industrial scale productions are being actively pursued by our group.

Experimental

Large-scale reactions with O_2 , air/ O_2 , or air

Synthesis of **2a with an O_2 balloon.** A 250 mL flask was degassed to remove the air inside completely and refilled with O_2 by a balloon of O_2 three times. Then $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (182.3 mg, 0.75 mmol), TEMPO (120.3 mg, 0.75 mmol), and **1a** (5.6549 g, 30 mmol) in MeCN (30 mL) were added sequentially. The resulting mixture was stirred at 25 °C for 16 h as monitored by TLC (petroleum ether/ethyl acetate = 5/1), filtered through a short column of silica gel [eluent: ethyl ether (120 mL)], concentrated under reduced pressure, and purified by column chromatography on silica gel [eluent: petroleum ether/ethyl ether = 30/1] to afford **2a** (5.4919 g, 98%) as an oil.

Synthesis of **2a with a bag of air with O_2 supplemented by a bag of pure O_2 .** To a 250 mL three-neck flask were added $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (968.2 mg, 4.0 mmol), TEMPO (636.7 mg, 4.0 mmol), and **1a** (7.5332 g, 40.0 mmol) in MeCN (80 mL) sequentially. Then a 42 L bag filled with air was connected to the flask. After stirring at 25 °C for 1.5 h, a 2 L bag of pure O_2 was connected to the flask to supplement O_2 . The resulting mixture was stirred at 25 °C for 4.5 h as monitored by TLC (petroleum ether/ethyl acetate = 5/1), filtered through a short column of silica gel [eluent: ethyl ether (150 mL)], concentrated under reduced pressure, and purified by column chromatography on silica gel [eluent: petroleum ether/ethyl ether = 30/1] to afford **2a** (7.4587 g, 100%) as an oil.

Synthesis of **2g with a slow flow of air.** To a 250 mL three-neck flask were added $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (967.8 mg, 4.0 mmol), TEMPO (639.4 mg, 4.0 mmol), and **1g** (5.5340 g, 40.0 mmol) in MeCN (120 mL) sequentially. A slow flow of air from the air cylinder (high purity, 30 mL min⁻¹) was then connected to the flask. The reaction was stirred at 25 °C for 46 h as monitored by TLC (petroleum ether/ethyl acetate = 5/1), filtered through a short column of silica gel [eluent: ethyl ether (150 mL)], concentrated under reduced pressure and purified by column chromatography on silica gel [eluent: petroleum ether/ethyl ether = 30/1] to afford **2g** (4.3224 g, 79%) as a white solid.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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